

1. Problem Statement

Consider the boundary value problem:

$$u_{xx} + u_{yy} = f(x, y), \quad (x, y) \in [-1, 1]^2, \quad (1)$$

$$u(x, y) = 0, \quad x, y \in \partial([-1, 1]^2). \quad (2)$$

We aim to solve the problem by a neural networks Green's function $\hat{G}$, the following two properties of Green's function may help us to achieve it.

1. The Green's function representation of $u(x, y)$,

$$u(x, y) = \int_{-1}^1 \int_{-1}^1 G(x, y, s, t) f(s, t) ds dt. \quad (3)$$

2. The adjoint problem of G ,

$$G_{ss} + G_{tt} = \delta(s - x)\delta(t - y), \quad (s, t) \in [-1, 1]^2, \quad (4)$$

$$G = 0, \quad (s, t) \in \partial([-1, 1]^2). \quad (5)$$

2. Neural Networks Structure

In this work, we aim to optimize the following loss function

$$\mathcal{L} = \frac{1}{M} \sum_{n=1}^M \left(\hat{G} * F_n - U_n \right)^2 + \lambda \left(\Delta(\hat{G} * F_n) - F_n \right)^2, \quad (6)$$

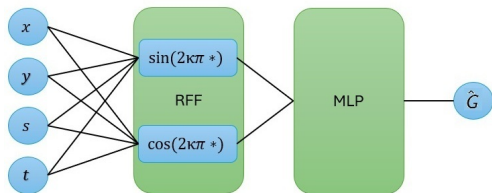
where $\hat{G}$ is the predict Green's function by NN, $*$ is the numerical integration, Δ is the Laplacian computed by differential matrix and λ is a parameter controlled the PDE loss. U_n and F_n are the value generated by

$$U_n = \sum_{(k_1, k_2) \in \Omega_K} \alpha_{k_1, k_2} \sin\left(\frac{k_1 \pi (X+1)}{2}\right) \sin\left(\frac{k_2 \pi (Y+1)}{2}\right), \quad (7)$$

$$F_n = \sum_{(k_1, k_2) \in \Omega_K} \alpha_{k_1, k_2} \left(-\left(\frac{k_1 \pi}{2}\right)^2 - \left(\frac{k_2 \pi}{2}\right)^2 \right) \sin\left(\frac{k_1 \pi (X+1)}{2}\right) \sin\left(\frac{k_2 \pi (Y+1)}{2}\right), \quad (8)$$

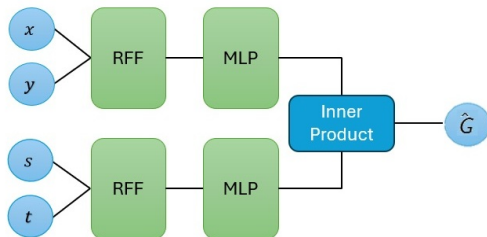
where Ω_K is a finite set of frequency pairs (k_1, k_2) randomly sampled from $\mathbb{N}^2$ for each sample n , and α_{k_1, k_2} represents the randomly generated coefficients and X, Y are 2D grid points.

Plan A:



- Input: $(x, y, s, t) \Rightarrow N^4$
- Size of training data: $M \cdot N^4$

Plan B:



- Input: $(x, y) \ \& \ (s, t) \Rightarrow 2N^2$
- Size of training data: $M \cdot 2N^2$

Since the size of the input vector determines the training time, we try to decrease it. We employ two separate networks and compute the inner product of their output vectors, which mimics the separation of variables in Plan B.

3. Experiment

In this work, we choose the following function as the test,

$$u(x, y) = \sin(\pi x) \sin(\pi y), \quad f(x, y) = -2\pi^2 \sin(\pi x) \sin(\pi y). \quad (9)$$

Here are other settings of the neural network,

- $N = 22$
- $M = 5$
- Activation: GELU
- Optimizer: Adam

RFF:

- FFsize: 100
- $\sigma = 1$
- $\mu = 0$

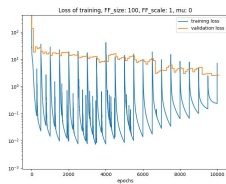
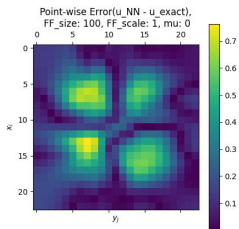
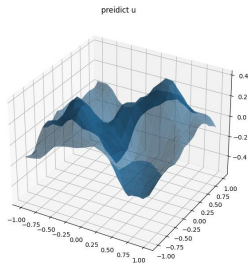
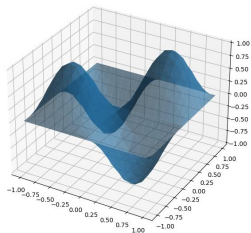
MLP:

- Plan A: [200, 50, 50, 1]
- Plan B: [200, 50, 50, 50]

and we measure the sup norm $\|L\|_\infty = \max_{ij} |u_{ij} - v_{ij}|$ between the exact u and the approximation v by computing the numerical integration * of the predict Green's function $\hat{G}$ and the exact f on the grid point (x_i, y_j) .

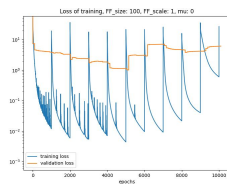
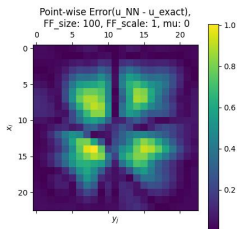
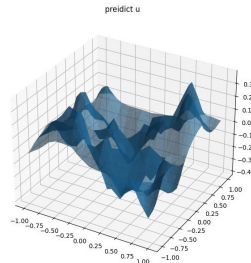
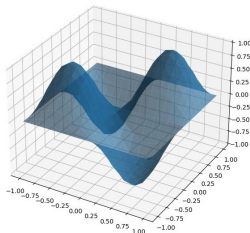
Plan A:

- Number of trainable parameters: 12651
- Training time: 233m 16s
- $\|L\|_{\infty} = 7.63e - 01$



Plan B:

- Number of trainable parameters: 30300
- Training time: 5m 37s
- $\|L\|_{\infty} = 8.48e - 01$



In another experiment, we aim to optimize the following loss function

$$\mathcal{L} = \frac{1}{M} \sum_{n=1}^M \left(\hat{G} * F_n - U_n \right)^2 + \lambda \left(\Delta(\hat{G} * F_n) - F_n \right)^2. \quad (10)$$

Here, we fix U_n and F_n to the value generated by the test function

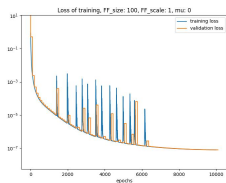
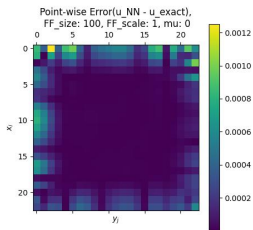
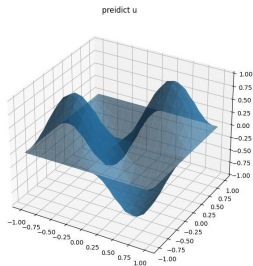
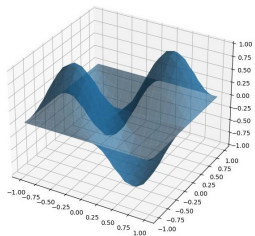
$$U_n = \sin(\pi X) \sin(\pi Y), \quad (11)$$

$$F_n = -2\pi^2 \sin(\pi X) \sin(\pi Y). \quad (12)$$

The other parameters are as same as the previous. By this experiment, we can validate whether the loss function works.

Plan A:

- Number of trainable parameters: 12651
- Training time: 234m 25s
- $\|L\|_{\infty}^{\text{exact } u} = 1.25e - 03$



Plan B:

- Number of trainable parameters: 30300
- Training time: 5m 33s
- $\|L\|_{\infty}^{\text{exact } u} = 3.12e - 04$

